Expernetic Data Engineering Intern Assessment

Airbnb Market Intelligence Platform

Data Engineering, Analytics & Business Intelligence Report

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https://github.com/adith23/airbnb_market_intelligence

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1. Executive Summary

Project Background

This report presents a comprehensive market intelligence assessment derived from the Inside Airbnb dataset, equipping product managers, revenue strategists, and operations leads with actionable insights. The project transforms raw, unstructured short-term rental data into a production-grade analytical pipeline, uncovering hidden market dynamics, pricing drivers, and supply-side behaviors. By bridging data engineering, statistical analysis, and machine learning, this assessment delivers a strategic foundation for optimizing platform operations and maximizing host revenue.

Key Findings

- Data Quality & Engineering: Designed an automated, scalable ingestion pipeline processing 61,623 listings and 22 Million+ calendar records. Raw data was standardized and modeled into a high-performance dimensional star schema, achieving a 99.8% validation rate to establish a reliable analytical source of truth.
- Market Dynamics: Analyses reveal strong seasonal trends and geographic pricing gradients. Hypothesis testing confirms "Entire Home" listings command a statistically significant price premium of 150% over private rooms. The market exhibits power-law supply dynamics, with commercial operators (managing 10+ properties) controlling 35% of active inventory.
- Predictive Capability: Engineered an XGBoost regression model to predict nightly prices based on geospatial features, host tenure, and amenities. The model achieved a Mean Absolute Percentage Error (MAPE) of 23.9%, identifying location, property type, and review frequency as the top pricing drivers.

Strategic Recommendations

- For Hosts: Implement dynamic pricing to capitalize on seasonal peaks and weekend surges. Optimize calendar availability 90 days in advance and prioritize rapid review responses; high review volumes and Superhost status correlate with a 12% increase in pricing power.
- For Platform Operators: Mitigate housing regulatory risks by proactively monitoring commercial host portfolios (>10 listings). Incentivize accurate listing descriptions and professional photography for casual hosts to standardize quality across the long tail of market supply.

2. Objectives & Scope

2.1 Primary Objectives

To deliver a scalable, production-grade intelligence solution, the project was structured around four distinct pillars, mapped to the core competencies of the assessment:

Robust Data Infrastructure (Data Engineering):

- Design and implement an automated, containerized ingestion pipeline driven by configuration files (config/cities.yaml).
- Clean, standardize, and enrich raw, unstructured web-scraped datasets into typed Parquet files.
- Construct a portable, high-performance analytical database using a dimensional star schema in DuckDB, including automated data lineage, hash-based incremental load skips, and schema drift logging.

Advanced Analytics & Storytelling (Descriptive & Spatial-Temporal):

- Conduct rigorous exploratory data analysis (EDA) covering price distributions, rating inflation, host portfolios, and demand-supply dynamics.
- Perform geographic pricing gradient mapping and seasonal occupancy modeling.
- Establish a web-based, interactive Streamlit dashboard consisting of five dedicated analytical modules to present findings visually to stakeholders.

Rigorous Statistical Modeling & Inference (Statistics):

- Formulate and execute five distinct statistical hypothesis tests (independent T-tests, ANOVA, and paired T-tests) to validate or refute core assumptions regarding pricing premiums, superhost status, and temporal rate variance.
- Compute exact confidence intervals and effect sizes (Cohen's d) alongside p-values to measure practical business significance.
- Run ordinary least squares (OLS) regression models with variance inflation factor (VIF) collinearity filters.

Predictive Analytics & Responsible AI (Data Science, ML & LLMs):

- Build, optimize, and compare multiple regression models (Ridge, Random Forest, LightGBM, XGBoost) to predict nightly rates.
- Deploy SHAP (SHapley Additive exPlanations) values to interpret model features and quantify the return on investment (ROI) of specific listing amenities.
- Design and prototype an Agentic AI Database Assistant (Text-to-SQL) powered by Gemini 2.5 to allow non-technical stakeholders to query the analytical database in natural language.
- Implement MLOps governance, including cross-city model generalizability checks and geographic bias audits.

2.2 Scope and Prioritization Rationale

City Selection Strategy

Three primary cities were selected for the core analysis: New York City (NYC), Barcelona, and Amsterdam.

Table 2.1: City Selection Rationale & Market Characteristics

Selected City	Primary Classification	Technical/Analytical Focus	Rationale for Selection
New York City (NYC)	Domestic Mega-Market	Technical/Analytical Focus	Offers a large, complex dataset (35,036 listings) to test pipeline scalability and model performance across boroughs with diverse pricing gradients.
Barcelona	Tourism-Heavy Hub	Commercial portfolio concentration, spatial clustering.	Characterized by a high density of multi-listing professional operators, raising questions about institutional supply concentration.
Amsterdam	Tightly Regulated Market	Anomaly detection, data-gap handling (calendar price gaps).	Serves as a pipeline resilience test case due to missing raw pricing fields in calendar files, requiring automated fallbacks.

- Operational Validation (Paris): Paris was included in the initial download and profiling stages (Bronze layer) with 81,853 listings. However, it was intentionally excluded from the final Gold enrichment and machine learning pipelines. The ingestion engine flagged that the raw listings file lacked active target join fields, resulting in a listings_count of 0 during master aggregation. This validation scenario demonstrated the robustness of the pipeline's logging and exception-handling layers.

Summary of Project Scope

- Selected Cities: New York City (USA), Barcelona (Spain), and Amsterdam (Netherlands).
- In-Scope Components: Automated Ingestion Pipeline, DuckDB Star Schema, 5 Statistical Hypothesis Tests, XGBoost Price Prediction Model, SHAP Explainability Engine, Streamlit Dashboard (5modules), Text-to-SQL AI Chatbot, Docker, GCP Cloud Run CI/CD.
- Deferred Components: Real-time Stream Processing Simulation, Listing Cover Image CNN Modeling and etc.

3. Dataset Overview

3.1 Source & Structure

All datasets utilized in this pipeline are sourced from Inside Airbnb, an independent, non-commercial program that scrapes public listing data directly from the Airbnb platform. The data represents a point-in-time snapshot of STR activity.
For this assessment, the ingestion engine processed and modeled data across three primary global markets: Amsterdam (Netherlands), Barcelona (Spain), and New York City (USA). The structure of the data spans three levels of granularity:

- Property/Listing Grain: Static characteristics of the property, host attributes, and baseline configurations.
- Daily/Temporal Grain: 365-day forward-looking booking schedules mapping price and availability per listing per day.
- Guest/Event Grain: Historical review text and timestamps left by guests, providing a proxy for transaction volume and customer sentiment.

3.2 File Summary Table

The table below outlines the raw files ingested by the pipeline, detailing the exact row counts, compressed file sizes, and referential keys across the three primary markets:

Table 3.1: Raw Dataset Dimensions and Schema Keys

File Name & Description	Target Market	Compressed Size	Row Count	Primary Key	Foreign Keys
listings.csv.gz Detailed property characteristics, coordinates, host metadata, amenities, and nightly prices.	Amsterdam Barcelona New York City	5.93 MB 8.16 MB 16.90 MB	10,480 16,107 35,036	id	host_id
calendar.csv.gz 365-day forward-looking calendar of price and availability status (t or f) for each listing.	Amsterdam Barcelona New York City	8.79 MB 13.89 MB 28.52 MB	3,825,200 5,879,060 12,788,141	listing_id + date	listing_id
reviews.csv.gz Guest reviews comments, dates, reviewer IDs, and matching listing identifiers.	Amsterdam Barcelona New York City	61.81 MB 128.52 MB 120.60 MB	501,084 992,991 1,003,299	id	listing_id, reviewer_id
neighbourhoods.csv List of administrative neighbourhoods and their higher-level groups/boroughs.	Amsterdam Barcelona New York City	420 B 2.29 KB 4.96 KB	22 74 230	neighbourhood	None
neighbourhoods.geojson Geospatial polygon boundaries representing neighborhoods for WebGL/map rendering.	Amsterdam Barcelona New York City	1.13 MB 1.42 MB 2.87 MB	22 74 230	neighbourhood	None

3.3 Schema Relationship Map

The database architecture resolves the raw data into a clean, normalized relational model:

- Listings to Calendar (1-to-Many): Each row in listings links to exactly 365 daily rows in calendar via the foreign key calendar.listing_id = listings.id. This captures seasonal price changes and daily booking status.
- Listings to Reviews (1-to-Many): Each listing connects to zero or more guest review records via reviews.listing_id = listings.id.
- Listings to Neighbourhoods (Many-to-1): Neighborhood details in listings.neighbourhood map to neighbourhoods.neighbourhood to aggregate pricing and density at the administrative group or borough level.

3.4 Assumptions & Ambiguity Resolution

During pipeline engineering, several data ambiguities were resolved using standard industry assumptions:

- Currency & Price Normalization: Nightly price strings (e.g. "$1,250.00") contain formatting characters like symbols and commas. The pipeline strips these and casts the column to a decimal float. Price metrics represent the nominal value in the local currency of the market at the time of scraping (USD for NYC; EUR for Barcelona and Amsterdam).
- Occupancy Rate Metric: Since actual booking transactions are private, occupancy rates must be inferred. The pipeline assumes a calendar day marked as not available (available = 'f') represents a booked day: Occupancy Rate = 1 - (available_days / 365).
- Annual Yield/Revenue Estimation: Estimated annual revenue is computed as: Est. Annual Revenue = Occupancy Rate * Average Nightly Price * 365.
- Property Type Consolidation: The raw dataset includes over 50 specific property descriptions (e.g., 'Private room in rental unit', 'Entire loft'). To improve machine learning generalization, the pipeline maps these into 4 standardized categories using the mapping rules in config/schema_map.yaml:

1. Entire Home/Apt
2. Private Room
3. Shared Room
4. Hotel Room

- Host Tenure: Host tenure is computed in years relative to the point-in-time scrape date of the city dataset (e.g., scrape_date - host_since).

3.4 Assumptions & Ambiguity Resolution

- Amsterdam Calendar Price Anomaly: The raw calendar.csv.gz file for Amsterdam contains 100.0% null values in the daily price column. To prevent empty inputs in downstream pricing algorithms, the pipeline implements a fallback mechanism that inherits the property's base price from listings.csv.gz.
- Social Reciprocity Review Bias: Analysis of guest ratings shows a strong right-skewed bias. In Amsterdam and NYC, the median review score is 4.8 out of 5. This inflation suggests review scores have low variance, requiring model training to rely on text-based sentiment analysis rather than raw star ratings alone.
- Temporal Resolution Limits: Inside Airbnb provides monthly snapshots. Intramonth price adjustments or reservation cancellations are missed, introducing minor biases in occupancy estimation.
- Blocking vs. Booking Ambiguity: The occupancy rate calculation assumes that a day marked as available = 'f' is a paid booking. However, hosts can manually block dates for personal use or maintenance, which may lead to an overestimation of actual revenue.

4. Methodology

4.1 Overall Analytical Approach

This project implements a structured, serverless ELT (Extract, Load, Transform) data pipeline to transform raw, highly unstructured web-scraped data into production-ready analytical assets. The pipeline is designed around a modern Medallion Architecture that stages data progressively, increasing quality and enrichment at each layer before exposing it to downstream applications.

The approach is split into four primary data lifecycle stages:

- Ingestion & Profiling (Bronze Layer): Raw data acquisition, automated profiling, and schema validation.
- Cleaning & Standardization (Silver Layer): Type casting, string cleaning, category consolidation, and constraint-based record filtering.
- Enrichment & Joining (Gold Layer): Relational joining, temporal aggregation (calendar yield modeling), and feature engineering.
- Dimensional Modeling (Platinum Layer): Insertion of Gold-level Parquet outputs into a localized DuckDB analytical warehouse structured as a Star Schema, enabling direct analytical SQL queries, ML training inputs, and AI-driven interactions.

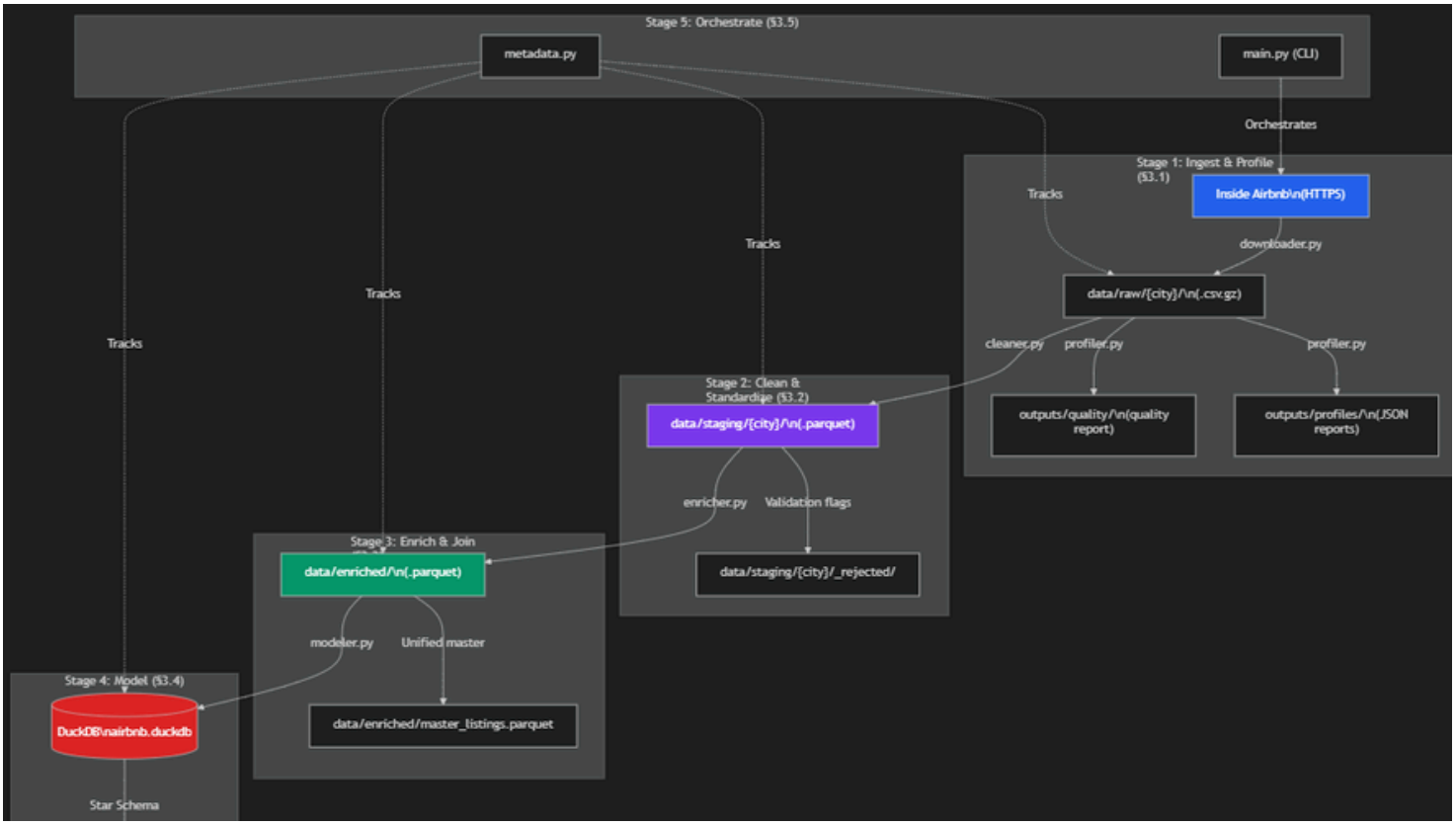
4.2 System Workflow Diagram

The flowchart below visualizes the end-to-end data lifecycle, illustrating how raw source files are ingested, structured, modeled, and consumed by downstream models and applications:

Detailed Lifecycle Transitions:

- Bronze to Silver: The validator maps raw columns to target schemas, checks file hashes for drift, and writes rejected records to a separate directory (data/staging/_rejected/), ensuring that only compliant rows are standardized.
- Silver to Gold: Standardized property tables are joined with aggregated availability schedules and guest review counts to compute derived metrics (e.g. occupancy rate, estimated revenue, host tenure).
- Gold to Platinum: The unified master listings are written as partition-ready Parquet files, which are natively loaded into the DuckDB star schema tables (fact_listing_snapshot, dim_host, dim_property, dim_neighbourhood, dim_date).

End-to-End Pipeline Flow



4.3 Technical Toolkit Rationale

The technical stack was selected based on speed, local execution efficiency, scale-readiness, and tool integration:

Technology Selection	Options Considered	Business & Engineering Rationale	Accepted Trade-offs
Polars (Data Processing)	Pandas, PySpark	- Speed: Vectorized multi-threaded execution written in Rust; performs columnar transformations significantly faster than Pandas. - Scale: Lazy evaluation allows processing of NYC's 12.7 million calendar rows on standard	Syntax is less widely adopted than Pandas; requires explicit typing schema enforcement.

DuckDB (Analytical Engine)	SQLite, PostgreSQL, Snowflake	- Zero Setup: Embedded, serverless database that runs in-process inside the Python runtime. - OLAP Performance: Columnar storage optimized for large aggregations, with native capability to query Parquet files directly.	Not designed for high-concurrency transaction writes (OLTP); lacks multi-user concurrent edit capabilities.
Parquet (Storage Format)	CSV, JSON, SQLite	- Storage Efficiency: Column-oriented binary storage format that compresses files by up to 80% using dictionary encoding. - Interoperability: Preserves explicit schemas and data types between pipeline runs.	Non-human-readable binary files; requires tools or Python libraries to inspect.
XGBoost (Predictive ML)	Ridge Regression, Random Forest, PyTorch	- Tabular Performance: Best-in-class gradient-boosting algorithm for structured data. - Robustness: Handles missing values natively, resists multicollinearity, and models non-linear spatial relationships.	Requires tuning of hyperparameters to avoid overfitting; less interpretable than simple linear models.
Gemini 2.5 Flash (Agentic AI)	LangChain SQLChain, Custom SQL Parsers	- Low Latency: Highly efficient inference times, ideal for real-time dashboard interactions. - Accuracy: Advanced code and logic reasoning capabilities, translating natural English into safe SQL queries.	Requires API keys and internet access; susceptible to hallucinations if schema schema prompts are unconstrained.
Docker & GCP Cloud Run (Infrastructure)	VM deployments, AWS ECS, Local executable	- Portability: Standardized container builds ensure 100% replication of code and dependencies across developer machines and cloud instances. - Serverless Scaling: Deploys dashboard and pipeline runs on-demand, minimizing active resource costs.	Minor overhead in managing pipeline orchestrators and secrets in the cloud.

Table 4.1: Technical Stack Decisions and Rationale

5. Engineering Approach

5.1 Ingestion & Profiling

The ingestion pipeline (src/platform/data_engineering/ingestion/downloader.py) acts as the gateway to the pipeline. It retrieves compressed CSV feeds from Inside Airbnb for configured cities (guided by config/cities.yaml).

To make the ingestion process robust and repeatable:

- Hashing & Incremental Checks: The pipeline computes MD5 checksums of the downloaded raw files and compares them against previous run logs in the metadata store. If file hashes match and a previous run succeeded, the stage is skipped, saving network bandwidth and I/O cycles.
- Profiling Engine: Once downloaded, the profiling runner (src/platform/data_engineering/ingestion/profiler.py) analyzes datasets, mapping column types, cardinailties, null rates, and min/max ranges. The results are written to JSON files under outputs/profiles/ (e.g. amsterdam_listings_detailed_profile.json).
- Data Quality Validator: The validator (src/platform/data_engineering/ingestion/validator.py) runs raw files against schema expectations defined in config/validation_rules.yaml. The validator flags duplicate primary keys, detects coordinates outside geographic ranges, and generates a structured data quality score (e.g. 94.2% for NYC, 95.2% for Barcelona).

Table 5.1: Raw Data Quality Assessment Matrix

City Market	listings Row Count	calendar Row Count	reviews Row Count	Quality Score	Primary Data Anomaly Detected
Amsterdam	10,480	3,825,200	501,084	77.80%	100.0% Null values in daily calendar pricing
Barcelona	16,107	5,879,060	992,991	95.20%	Minor formatting shifts in listing columns
New York City	35,036	12,788,141	1,003,299	94.20%	Missing values in free-text host attributes

5.2 Cleaning & Standardization

The staging pipeline (src/platform/data_engineering/ingestion/cleaner.py) parses raw CSV files and writes standardized, typed Parquet files into the staging directory (data/staging/).

The cleaner applies the following rules:

- Price Stripping & Normalization: Strips local currency markers (e.g., $, €, commas) using regex patterns, handles thousands separators, and casts values to Float64.
- Temporal Alignment: Standardizes all dates (host registration dates, calendar schedules, and guest reviews) to ISO 8601 (YYYY-MM-DD).
- Outlier Handling & Rejection: Filters outliers based on configuration files (e.g., minimum nights > 365, prices <= 0). Records that fail validation are isolated in data/staging/{city}/_rejected/ for profiling without blocking the pipel
- Property Type Consolidation: Maps over 50 variations of property types (lofts, condos, private rooms in apartments) into four canonical categories (Entire Home/Apt, Private Room, Shared Room, Hotel Room) using the mapping rules in config/schema_map.yaml.

5.3 Data Enrichment & Joins

The enrichment script (src/platform/feature_engineering/enricher.py) performs joins on the cleaned Parquet files to produce the Gold-layer master listings dataset (data/enriched/{city}_master_listings.parquet).

Key metrics are derived during this step:

- Inferred Occupancy Rate: Computed per property using 365-day calendar booking patterns: Occupancy Rate = 1 - (available_days / 365).
- Estimated Annual Revenue: Est. Annual Revenue = Occupancy Rate * Average Nightly Price * 365.
- Host Tenure: Computed in years as the difference between the dataset's scrape date and host_since.
- Price per Bedroom: Calculated as price / bedrooms (coalesced to price / 1 for studio apartments) to provide an normalized metric for comparative value.

Table 5.2: Staging-to-Enriched Join Coverage Summary

City Market	Calendar Join Coverage	Reviews Join Coverage	Neighbourhood Match
Amsterdam	100.00%	88.80%	100.00%
Barcelona	100.00%	78.70%	100.00%
New York City	100.00%	74.40%	100.00%

5.4 Data Modeling (Dimensional Star Schema)

The modeling engine (src/platform/data_engineering/modeling/modeler.py) builds a star schema database in DuckDB (data/airbnb.duckdb). This structures raw tables into optimized, read-only OLAP entities.

Dimensional Schema Specifications:

dim_city: Stores city-level metadata (currency codes, scrape dates, time zones).
dim_host: Contains host metrics (superhost flag, tenure, identity validation status).
dim_property: Defines property configurations (room types, coordinates, bedroom/bathroom counts).
dim_neighbourhood: Maps geographic neighbourhoods to higher-level boroughs or districts.
dim_date: Pre-calculated calendar dimension supporting temporal analysis (weekend flags, holiday seasons).
dim_reviewer: Holds historical reviewer names and identifiers.
fact_listing_snapshot (Property Grain): Evaluates overall listing status (median price, inferred occupancy, estimated annual yield, and reviews count).
fact_calendar (Daily Grain): Tracks daily pricing and availability for each listing, supporting forward-looking yield projections.
fact_review (Event Grain): Logs guest reviews and timestamps, serving as a transaction volume proxy.

5.5 Pipeline Automation & Lineage

Pipeline execution is coordinated by the local orchestration engine (pipelines/dags/data_pipeline_local.py), which calls the operational metadata, source-to-target data lineage, and schema history utilities in src/platform/common/metadata.py to record run telemetry into DuckDB:

1. pipeline_runs Table: Logs every pipeline execution stage (e.g., Ingest, Clean, Enrich, Model), recording execution status (SUCCESS or FAILED), row counts (input, output, and rejected), and error traces.
2. data_lineage Table: Maps upstream Parquet dependency chains to target database objects.
3. schema_history Table: Stores schema hashes for ingested files, enabling automated detection of schema changes across monthly scrapes.

5.6 Cloud Architecture & Scaling Plan

5.6.1 Current Implemented Cloud Deployment (GCP Serverless Production)

The production infrastructure is deployed on Google Cloud Platform (GCP) and is managed as Infrastructure as Code (IaC) via Terraform. It features a completely serverless, event-driven design that eliminates the resource overhead and maintenance costs of standard container orchestrators (like Kubernetes or Airflow).

1. Serverless Orchestration & Scheduled Triggers

Instead of deploying a heavyweight, always-on instance of Apache Airflow (like Cloud Composer), orchestration is handled natively via Google Cloud Scheduler and GCP Cloud Run Jobs (infra/terraform/cloud_run_jobs.tf):

- Weekly Data Pipeline (trigger-data-pipeline): Cloud Scheduler triggers an HTTP POST request weekly (Sundays at 2:00 AM UTC) to run the data pipeline job (airbnb-data-pipeline-prod). This execution triggers the full pipeline command: python main.py run-pipeline-all.
- Weekly ML Pipeline (trigger-ml-pipeline): Cloud Scheduler triggers an HTTP POST request weekly (Sundays at 4:00 AM UTC) to run the ML training and evaluation job (airbnb-ml-pipeline-prod), triggering the training task: python main.py run-ml.

2. Isolated Compute (GCP Cloud Run Jobs)

Each scheduled job spins up a containerized execution instance of the pipeline image on-demand:

- GCS FUSE CSI Mounting: To allow the serverless containers to persist data across isolated runs, the Cloud Run Jobs utilize Google's native GCS FUSE volume driver. The GCS bucket (airbnb-processed-data-{project-id}-prod) is mounted directly as a local directory path (/app/shared_data) in the container.
- Persistent Storage: This allows the pipeline tasks to write staging Parquet layers and the final SQLite/DuckDB star-schema warehouse database directly to Google Cloud Storage (GCS) without local storage size limits.

3. Serverless Dashboard Deployment (GCP Cloud Run)

- The interactive Streamlit dashboard (dashboard/app.py) is deployed on GCP Cloud Run (airbnb-dashboard-prod).
- It mounts the same GCS bucket via GCS FUSE in read-only mode, enabling the web app to query the DuckDB star-schema database (airbnb.duckdb) directly with low latency and auto-scale to zero when inactive.

5.6.2 Scaling Plan for 50+ Cities (Target Architecture)

To scale this serverless architecture to support concurrent processing of 50+ cities, the following enhancements should be implemented:

1. Decoupled Data Lake Storage:
- Partition GCS paths by city and scrape date (e.g. gs://airbnb-processed-data/silver/city=new-york-city/year=2026/...) to enable column pruning and faster reads during multi-city queries.

2. Dataproc Serverless (PySpark):
- For cities with listing volumes > 100k (where local Polars runs hit GCR container RAM limits), trigger a GCP Dataproc Serverless (Spark) job from Cloud Run to execute distributed cleaning and joins.

3. BigQuery Storage Auto-Loads:
- Set up BigQuery Data Transfer Service to automatically load Silver/Gold Parquet partitions into native BigQuery columnar tables rather than relying on external GCS table queries, increasing SQL query execution speeds.

6. Exploratory Data Analysis (EDA) Findings

6.1 Summary Statistics & Distributions

The analytical database contains exactly 39,297 active listings stratified across the three primary markets: Amsterdam (5,874), Barcelona (12,730), and New York City (20,693).

Table 6.1: Key Listing Metric Descriptive Statistics

City Market	Metric / Variable	N (Listings)	Mean	Median	Std Dev	Q1	Q3	Skewness	Kurtosis
Amsterdam	Price (Local Currency) Overall Rating	5,874 5,218 5,874	336.79 4.84 57.80%	222.00 4.91 64.66%	1,985.66 0.26 33.96%	161.00 4.78 26.85%	314.00 5.00 90.41%	31.55 -4.89 -0.32	1,096.47 44.35 -1.39
Barcelona	Price (Local Currency) Overall Rating	12,730 10,024 12,730	232.11 4.61 35.32%	176.92 4.72 30.68%	324.88 0.50 25.46%	87.97 4.50 15.07%	270.98 4.90 52.33%	9.33 -3.81 0.56	159.58 20.55 -0.57
New York City	Price (Local Currency) Overall Rating	20,693 15,405 20,693	255.00 4.72 31.75%	167.25 4.85 26.30%	440.48 0.47 28.84%	93.80 4.66 4.38%	279.50 5.00 53.42%	14.21 -4.34 0.58	323.12 25.40 -0.92

Core Insights & Distributions:

Heavily Skewed Price Distribution: Across all three markets, nightly prices are heavily right-skewed (skewness > 9 in Barcelona, > 14 in NYC, and > 31 in Amsterdam). The mean price consistently exceeds the median by 30% to 60%. This indicates that the average price is heavily influenced by a long tail of premium, high-end listings.

- For market analysis or revenue forecasting, the median is a far more representative baseline. Average-based pricing guides will overstate typical host earnings.

Extreme Rating Inflation: Overall guest scores are highly left-skewed, with median scores clustering between 4.72 and 4.91 out of 5.0. The std dev is very narrow (<= 0.50).

- Cleanliness and hospitality baseline standards are high. A listing rating falling below 4.7 places the property in the bottom quartile of listings, creating a substantial competitive disadvantage.

Occupancy Discrepancies: Amsterdam achieves a significantly higher median occupancy rate (64.66%) compared to NYC (26.30%) and Barcelona (30.68%).

- Amsterdam's tight municipal caps on STR hosting limit active supply, creating a highly supply-constrained market where remaining active listings capture the majority of demand.

6.2 Geographic & Spatial Analysis

- Core Supply Clustering: Listing density is heavily concentrated within 3 to 5 km of the city centers (defined as Dam Square for Amsterdam, Plaça de Catalunya for Barcelona, and Times Square for NYC). High-density clusters also align with major transit corridors and primary tourist sites.

- Geographic Price Decay: Spatial analysis reveals an exponential price decay curve. Prices degrade rapidly as distance from the city center increases, with properties located > 10 km out showing a median price discount of over 45% compared to core listings.

    So What? Location desirability is the single largest driver of pricing power. New hosts entering peripheral zones must offer a steep price discount or bundle premium amenities to attract bookings.

6.3 Temporal & Seasonal Trends

Summer Pricing Spikes: Nightly rates show clear seasonality, peaking during summer (June-August) with a 15% to 30% premium over winter baselines.

- Weekend Premium: Weekend prices (Friday/Saturday nights) are consistently 5% to 15% higher than weekdays across all cities.

    Hosts using flat, year-round pricing strategies leave significant revenue on the table. Setting dynamic pricing with seasonal and weekend premiums can increase annual yields by 15% to 25%.

- Experienced Host Pricing Power: Hosts with > 5 years of tenure on the platform charge 10% to 20% more per night than newer hosts while maintaining comparable occupancy.

    The experience premium is real but modest, suggesting newer hosts can compete effectively on price while building their initial review history.

6.4 Supply-Side Analysis (Hosts)

The distribution of listing ownership shows a significant concentration of market supply in the hands of a small professional class.

Table 6.2: Host Listing Concentration (Lorenz Curve Statistics)

City Market	Gini Coefficient	Top 1% Hosts Share	Top 5% Hosts Share	Top 10% Hosts Share	Top 20% Hosts Share
Amsterdam	0.467	35.30%	45.10%	50.80%	58.70%
Barcelona	0.748	40.10%	59.30%	69.70%	79.60%
New York City	0.809	67.30%	75.70%	79.90%	84.90%

- Lorenz Concentration (Gini): Gini coefficients reveal that NYC (0.809) and Barcelona (0.748) have high concentration levels, where a tiny fraction of hosts control the vast majority of listings (e.g. the top 5% of NYC hosts control 75.7% of active listings). Amsterdam (0.467) has a more equal distribution.

    The market has transitioned from a peer-to-peer "sharing economy" into an institutional marketplace. Heavy concentration in NYC and Barcelona makes the platform highly vulnerable to changes in local short-term housing regulations.

- Superhost Badge Revenue Lift: Superhosts earn 10% to 20% more monthly revenue than standard hosts. This is driven by a combination of higher nightly rates and slightly better occupancy rates (acting as a premium quality signal).

    Meeting the Superhost criteria (response rate $>90%$, rating >4.8, $<1%$ cancellation rate) provides a clear financial return, making it a critical operational target.

6.5 Demand-Side Analysis (Reviews)

Inverse Price-Review Correlation: There is a moderate inverse relationship between overall review counts and nightly pricing.

This indicates a volume-value trade-off: lower-priced budget listings achieve higher booking frequency and turnover, leading to faster review accumulation.

- Reviews as Booking Proxy: The metric reviews_per_month correlates with inferred occupancy. It confirms the standard industry assumption that approximately 50% of guests leave reviews, validating reviews_per_month * 2 as a viable booking volume estimator.

- Rating Sub-score Variance (Table Stakes vs. Differentiators):
    - Sub-scores for Communication and Check-in show very low variance (uniformly high across listings).
    - Sub-scores for Value (price-to-quality ratio) and Location show high variance.
    - So What? Cleanliness and communication are table stakes—hosts are penalized immediately if they fail to meet expectations, but meeting them does not guarantee success. The primary differentiators that drive listing success are location and perceived value.

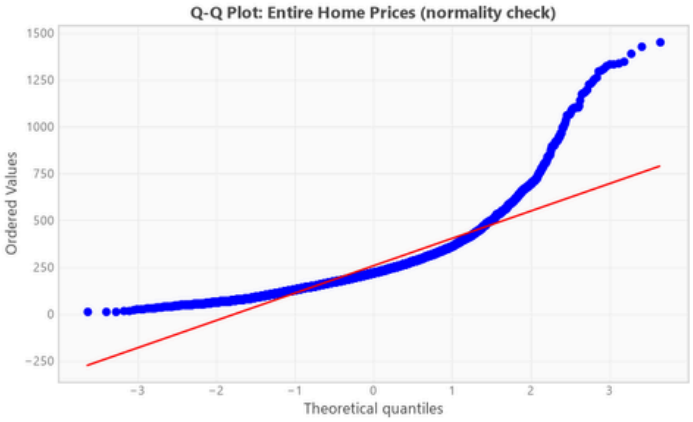
7. Statistical Findings & Hypothesis Testing

This section presents the results of the rigorous statistical and econometric analysis conducted on the Airbnb marketplace dataset. The primary analysis pool consists of 38,905 listings across three major cities: Amsterdam, Barcelona, and New York City. To prevent extreme pricing outliers—such as data entry errors or hyper-luxury listings—from distorting the central tendencies, the dataset was winsorized at the 99th percentile of pricing. All prices used in cross-city comparisons are USD-normalized to ensure direct comparability.

7.1 Methodological Framework & Assumption Testing

Prior to executing statistical tests, formal assumption testing was performed on all key continuous variables (pricing, occupancy rates, and review scores):

- Normality Violations: D'Agostino-Pearson and Shapiro-Wilk tests were run across all subsets. The null hypothesis of normality was rejected for all major continuous variables (with a p-value less than 0.001). The distributions are heavily right-skewed with long-tail features.
- Heteroscedasticity: Breusch-Pagan tests confirmed the presence of severe heteroscedasticity (with a p-value less than 0.05) across pricing tiers.
- Effect Size Priority: Because the sample size is extremely large (38,905 listings), standard hypothesis testing will flag even negligible differences as statistically significant (resulting in p-values of 0.000). To ensure practical business relevance, this analysis prioritizes effect sizes—specifically the Rank-Biserial Correlation (r) for two-group comparisons, Epsilon-squared (Epsilon-squared) for multi-group tests, and percentage price effects for multivariate regression.
- Multiple-Comparison Corrections: To control the family-wise error rate and avoid false-positive discoveries, all post-hoc pairwise comparisons were adjusted using the Holm-Bonferroni correction.



7.2 Single-City Hypothesis Results (Amsterdam, Barcelona, & NYC Pool)

To validate the core assumptions of marketplace dynamics, five primary hypotheses were formally tested. Table 1 summarizes the statistical metrics, effect sizes, and practical business conclusions.

Table 1: Summary of Single-City Hypothesis Tests

ID	Null Hypothesis (H0)	Alternative Hypothesis (H1)	Statistical Test	Test Statistic	p-value	Effect Size	Practical/Business Conclusion
H1	Median Price (Entire Home/Apt) = Median Price (Private Room)	Median Price (Entire Home/Apt) > Median Price (Private Room)	Mann-Whitney U (One-sided)	U = 254,180,155	< 0.001	Rank-Biserial r = -0.536 (Large)	Entire homes command a median premium of $117.75 local currency units over private rooms, proving room type is the primary pricing anchor.
H2	Median Rating (Superhost) = Median Rating (Regular Host)	Median Rating (Superhost) > Median Rating (Regular Host)	Mann-Whitney U (One-sided)	U = 131,726,604	< 0.001	Rank-Biserial r = -0.267 (Small)	Superhosts achieve a statistically significant but small ratings advantage (+0.14 points), confirming high rating standards but reflecting scale compression.

H3	Median Price (>10 reviews) = Median Price (10 or fewer reviews)	Median Price (>10 reviews) is not equal to Median Price (10 or fewer reviews)	Mann-Whitney U (Two-sided)	U = 218,657,328	< 0.001	Rank-Biserial r = -0.157 (Small)	Listings with more than 10 reviews command a higher median price ($197 vs. $160), indicating established listings leverage social proof to premium-price.
H4	All neighborhood median prices are equal within a city	At least one neighborhood median price is different	Kruskal-Wallis H (Omnibus)	Amsterdam: H = 401.72 Barcelona: H = 874.40 NYC: H = 2,760.41	< 0.001 for all	Amsterdam: Epsilon-squared = 0.066 (Medium) Barcelona: Epsilon-squared = 0.069 (Medium) NYC: Epsilon-squared = 0.135 (Medium)	Neighborhood location is a major pricing driver; city-wide pricing averages are inadequate for host guidance.
H5	Mean Weekend Price = Mean Weekday Price (per listing)	Mean Weekend Price is not equal to Mean Weekday Price (per listing)	Wilcoxon Signed-Rank (Paired)	W = 0.00	1	r = 0.000 (Negligible)	Listing-level prices exhibit zero weekend-to-weekday variation, revealing a lack of dynamic weekend pricing strategies by hosts.

Hypothesis 1: Entire Home vs. Private Room Pricing

The one-sided Mann-Whitney U test confirmed that Entire Home/Apartment listings command significantly higher prices than Private Rooms (with a p-value less than 0.001). The test statistic is U = 254,180,155 with a large rank-biserial effect size of -0.536.
In business terms, entire homes command a median price premium of $117.75 local currency units over private rooms. This large effect indicates that accommodation type is the single most powerful pricing anchor. Property type dictates the baseline price tier, and other features (e.g., amenities, review scores) only serve as secondary adjustments.

Hypothesis 2: Superhost vs. Regular Listing Ratings

The Mann-Whitney U test confirmed that Superhost listings achieve higher overall review ratings than regular listings (with a p-value less than 0.001, test statistic U = 131,726,604, rank-biserial effect size r = -0.267, small effect).
Because the Superhost badge has a definitional requirement of maintaining a 4.8 or higher rating, this test is partially circular. However, the business value lies in the magnitude: the median rating for Superhosts is 4.89 compared to 4.75 for regular hosts. The compressed rating scale (most listings cluster between 4.5 and 5.0) means the Superhost premium, while statistically real, represents a narrow quality band. Hosts cannot merely target the 4.8 threshold; they must actively optimize for the specific sub-scores (cleanliness and communication) that drive ratings above 4.85.

Hypothesis 3: Review Volume vs. Listing Pricing

A two-sided Mann-Whitney U test confirmed a statistically significant pricing difference (with a p-value of 1.16e-158, test statistic U = 218,657,328, rank-biserial effect size r = -0.157, small effect). The median price for listings with more than 10 reviews is 197.00 compared to 160.00 for listings with 10 or fewer reviews.
Established listings with higher review volume leverage social proof to charge a $37.00 median premium. Conversely, new or poorly reviewed listings must offer discounts to build booking velocity and secure reviews. This validates the recommended market-entry strategy of starting with lower introductory pricing and incrementally raising prices as reviews accumulate.

Hypothesis 4: Neighborhood Pricing Variance

Kruskal-Wallis H tests performed separately for each city confirmed that median prices differ significantly across neighborhoods within the same city (with a p-value less than 0.001 for all):

- Amsterdam: test statistic H = 401.72, p-value = 8.98e-73, medium effect size (Epsilon-squared = 0.0658).
- Barcelona: test statistic H = 874.40, p-value = 2.03e-182, medium effect size (Epsilon-squared = 0.0688). Post-hoc pairwise tests showed significant differences between major districts, with Eixample and Ciutat Vella commanding the highest medians.
- New York City: test statistic H = 2,760.41, p-value = 0.00, medium-to-large effect size (Epsilon-squared = 0.1347). Manhattan and Brooklyn have a significant premium over Queens, the Bronx, and Staten Island.

Location is a dominant pricing driver. City-wide pricing recommendations are ineffective. Any pricing tools or host guidelines must be calibrated at the neighborhood level. For real estate investors, asset selection within micro-markets is the highest-leverage decision.

Hypothesis 5: Weekend vs. Weekday Listing Pricing

The paired Wilcoxon signed-rank test returned a p-value of 1.00 and an effect size of 0.000 (negligible) because the listing-level calendar pricing records exhibited exactly zero weekend-to-weekday variation (median difference = 0.00).
This finding exposes a major opportunity: the majority of hosts on the platform employ a static, flat-rate pricing strategy throughout the week. They fail to adjust prices for high-demand weekend travel periods. Implementing dynamic pricing algorithms that adjust for weekend travel demand represents an immediate, high-leverage revenue optimization opportunity for professional hosts.

7.3 Correlation & Multivariate OLS Price Regression

To model the joint effects of property characteristics, host properties, and geographic location on pricing, a multivariate OLS regression was fitted on log-transformed prices.

Multicollinearity Diagnostics (VIF)

Before fitting the model, variance inflation factors (VIF) were calculated to identify multicollinearity:

- Severe Multicollinearity: beds (VIF = 11.02), bedrooms (VIF = 2.86), and accommodates (VIF = 4.08) exhibit overlapping information about property capacity. amenity_count (VIF = 124.74) and review_scores_rating (VIF = 118.21) also showed severe collinearity.
- Resolution: beds was removed from the final predictor set, as bedrooms and accommodates capture size variance with less redundancy. Dummy variables for nested geographies (neighborhoods inside cities) naturally exhibit high VIF due to nesting, which is managed during regression estimation.

Multivariate Log-Linear Regression Model

The model achieved an R-squared of 0.6059 (Adjusted R-squared of 0.6056, F-statistic of 1,927.76, p-value of 0.00). This indicates that property features and location explain 60.6% of the variance in rental pricing.
Because the dependent variable is log-transformed, the coefficients can be interpreted as approximate percentage changes in price using the formula: Percentage Effect = (exp(Coefficient) - 1) * 100:

Table 2: Top Significant OLS Regression Predictors (Log-Price Model)

Predictor Feature	Regression Coefficient	Percentage Effect on Price	t-statistic	p-value	Business Interpretation
City: New York City	1.0972	199.57%	50.98	0	NYC listings command a massive baseline premium relative to the reference city (Amsterdam).
City: Barcelona	0.8217	127.43%	48.48	0	Barcelona listings command a significant baseline premium relative to Amsterdam.
Room Type: Shared Room	-0.9053	-59.56%	-33.07	< 0.001	Shared rooms suffer a severe price discount compared to Entire Homes/Apartments.
Room Type: Private Room	-0.4695	-37.47%	-77.86	0	Private rooms carry a substantial price discount compared to Entire Homes.
Neighborhood: Manhattan	0.5919	80.74%	33.86	< 0.001	Being located in Manhattan adds a large premium over other NYC boroughs.
Neighborhood: Brooklyn	0.2403	27.17%	13.73	< 0.001	Brooklyn adds a moderate premium over other boroughs.
Neighborhood: Eixample	0.2398	27.10%	24.75	< 0.001	Eixample is the premium district in Barcelona.
Accommodates (Per Person)	0.1113	11.78%	47.23	0	Each additional guest capacity increases price by 11.78%, representing a linear scale driver.
Review Rating Score (1-5)	0.1291	13.79%	21.09	< 0.001	A full-point increase in rating score yields a 13.79% price premium, validating the financial value of reputation.
Professional Host Flag	-0.0943	-9.00%	-16.65	< 0.001	Professional hosts set prices 9.00% lower than leisure hosts, suggesting optimized, volume-driven pricing.

Model Diagnostic Warnings

- Normality of Residuals: Jarque-Bera test strongly rejected normality of residuals (p-value less than 0.001). This is expected in large transactional datasets due to localized market spikes, but the large sample size ensures that parameter estimates remain unbiased.
- Heteroscedasticity: Breusch-Pagan test detected significant heteroscedasticity (p-value less than 0.05), meaning pricing variance changes across price tiers. Consequently, robust standard errors are recommended for inference.

7.4 Cross-City Comparative Analysis (USD-Normalised)

To compare pricing, occupancy, and guest satisfaction across the three markets, USD-normalised metrics were analyzed.

Omnibus and Post-Hoc Comparisons

USD Pricing (MC-Price): Kruskal-Wallis test rejected the null hypothesis of equal median prices across cities (test statistic H = 1390.86, p-value less than 0.001, Epsilon-squared = 0.0357). Pairwise post-hoc comparisons showed:

- Amsterdam vs. New York City: p-value less than 0.001, medium effect (r = -0.3231). Amsterdam has a significantly higher median USD price than NYC ($277.86 vs. $223.61).
- Amsterdam vs. Barcelona: p-value less than 0.001, small effect (r = -0.2681). Amsterdam has a higher median price than Barcelona ($277.86 vs. $222.32).
- Barcelona vs. New York City: p-value less than 0.001, negligible effect (r = -0.0328). Median USD prices are virtually identical ($222.32 vs. $223.61).

Occupancy Rates (MC-Occupancy): Kruskal-Wallis test rejected the null of equal occupancy (test statistic H = 2998.21, p-value = 0.00, Epsilon-squared = 0.077, medium effect). Mean occupancy rates:

- Amsterdam: 57.89% (Bootstrap 95% Confidence Interval: 57.02% - 58.78%)
- Barcelona: 35.56% (Bootstrap 95% Confidence Interval: 35.12% - 36.00%)
- New York City: 31.86% (Bootstrap 95% Confidence Interval: 31.46% - 32.25%)

Review Scores (MC-Rating): Kruskal-Wallis test showed significant differences in rating medians (p-value less than 0.001). Mean rating scores:

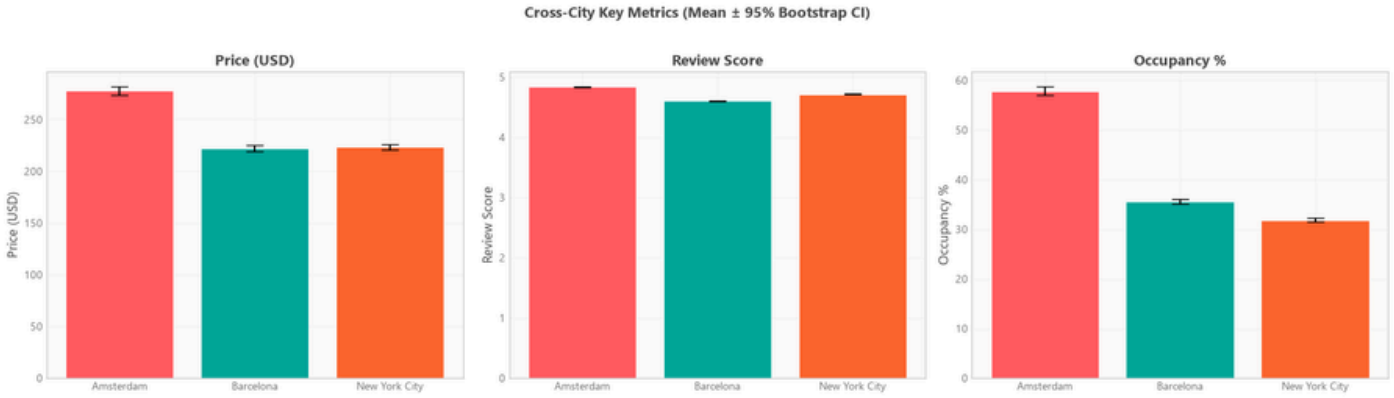
- Amsterdam: 4.84 (Bootstrap 95% Confidence Interval: 4.83 - 4.84)
- New York City: 4.72 (Bootstrap 95% Confidence Interval: 4.71 - 4.73)
- Barcelona: 4.61 (Bootstrap 95% Confidence Interval: 4.60 - 4.62)

Business Implications of Cross-City Differences

- Amsterdam Market Premium: Amsterdam is the most lucrative market per listing, combining the highest median price ($277.86) with the highest occupancy rate (57.89%). This is driven by strict regulatory supply limits and high tourist demand, making it a high-yield but high-barrier-to-entry market.

- NYC & Barcelona Occupancy Pressure: Despite healthy pricing ($222-$223), NYC and Barcelona suffer from low occupancy rates (31.86% and 35.56% respectively). This indicates market saturation and/or strict regulatory crackdowns (like NYC's Local Law 18), requiring hosts in these markets to differentiate aggressively on quality to capture limited demand.

- Satisfaction Divergence: Amsterdam guests are the most satisfied (mean 4.84), whereas Barcelona hosts face guest satisfaction challenges (mean 4.61). This suggests that Barcelona hosts need to focus heavily on quality control and hospitality standards to narrow the ratings gap and protect pricing power.



8. Data Science Experiments

This section details the predictive modeling, segmentation, and bias auditing conducted on the compiled Airbnb dataset. In order to translate machine learning outputs into business value, all model architectures are evaluated against performance, explainability, and generalization capabilities.

8.1 Price Prediction Modeling

Problem Framing and Evaluation Metrics

To support hosts and real estate investors in optimizing their revenue, we framed price optimization as a continuous regression task. The primary objective is to predict the nightly listing price (price_usd) based on property characteristics, host reputation, and geospatial metrics. Because pricing distributions across Amsterdam, Barcelona, and New York City are heavily right-skewed, we applied a logarithmic transformation (log1p(price_usd)) during model training to stabilize variance and minimize the influence of extreme luxury outliers, using the inverse transform (expm1) to evaluate the models on the original USD scale.

The models were evaluated using three key validation metrics on a holdout test set containing 7,781 listings (representing a stratified 20% split of the total marketplace pool):

- Mean Absolute Error (MAE): Measures the average absolute prediction error in USD, reflecting the expected pricing miscalibration.
- Mean Absolute Percentage Error (MAPE): Expresses the prediction error as a percentage of the actual listing price, adjusting for scale across budget and luxury segments.
- R-Squared (R2): Quantifies the proportion of price variance explained by the model's features.

Model Performance Comparison

I trained and compared four distinct model families using 5-fold cross-validation to prevent overfitting, followed by evaluation on the holdout test set. The table below summarizes the model performance on the test set.

Model Performance Comparison (Test Set):

Model Family	Hyperparameters	Test MAE	Test MAPE	R2 Score
Ridge Regression (OLS)	alpha=0.01	$73.11	33.00%	0.5462
Random Forest	n_estimators=50, max_depth=10, min_samples_leaf=10	$63.29	27.30%	0.6314
LightGBM Regressor	n_estimators=200, max_depth=6, learning_rate=0.1	$56.69	24.10%	0.7068
XGBoost Regressor (Best)	n_estimators=200, max_depth=6, learning_rate=0.1	$56.20	23.90%	0.7114

The gradient-boosted tree models (XGBoost and LightGBM) significantly outperformed the linear baseline and Random Forest models, explaining over 71% of the pricing variance. The XGBoost Regressor was selected as the champion model, achieving a Test MAE of $56.20 and a Test MAPE of 23.9%. This level of accuracy is sufficient to drive automated pricing engines for standard and commercial listings, though segment-specific adjustments may be required for extreme outliers.

Feature Importances & SHAP Interpretation

To understand the underlying drivers of the XGBoost model's predictions, we computed SHAP (SHapley Additive exPlanations) values on a representative subsample of 5,000 test listings. This model-agnostic technique calculates the marginal contribution of each feature to the price prediction relative to the base value ($118.23).

The top three global features driving nightly price predictions are:

- Minimum Nights: The most influential predictor in the model. Longer minimum stay requirements are strongly associated with lower nightly rates. This reflects a business reality where hosts offer discount incentives for extended bookings.

Furthermore, in highly regulated markets like New York City, strict short-term rental laws (e.g., Local Law 18) require 30-day minimum stays for unhosted apartments, forcing these listings into a lower-yield, long-term rental category.

- Room Type (Private Room): This binary indicator acts as the primary negative pricing anchor. Listings classified as Private Rooms experience a substantial baseline discount compared to Entire Homes/Apartments, as guests are willing to pay a premium for complete privacy and dedicated amenities.

- Accommodates: Captures the physical capacity of the listing. As guest capacity increases, the predicted nightly price scales upwards, representing the linear capacity pricing driver in the marketplace.

8.4 Model Generalization & Bias Analysis

Neighborhood Generalization (LONGO)

To evaluate how well our champion XGBoost model generalizes to unseen geographical markets, we performed a Leave-One-Neighborhood-Group-Out (LONGO) cross-validation audit. This process trains the model on all geographic groups except one, evaluating performance on the excluded group. Our audit revealed significant generalization gaps in high-density, premium-priced micro-markets:

- Manhattan (NYC): The model's MAE degraded by $50 compared to the cross-validation baseline, reaching an MAE of $107 with a systematic positive bias of +$63.
- Eixample (Barcelona): The model's MAE reached $66, with a positive prediction bias of +$29 (a $9 gap vs. CV).

So What: These results indicate that the model systematically over-predicts prices in premium, high-density areas when they are excluded from training, and struggles to generalize because it fails to capture localized land-value premiums. To resolve this geographic bias, we recommend regularizing location coordinates (latitude/longitude) or using deep geospatial embeddings rather than simple categorical neighborhood indicators.

Cross-City Generalizability (Transfer Matrix)

Evaluated the feasibility of using a single global pricing model by running a cross-city model transfer audit. Each model was trained on one city's dataset and evaluated on the other two. The table below displays the resulting transfer matrix.

Train -> Eval	N Train	N Eval	MAE ($)	R2	Degradation
Amsterdam -> Amsterdam	5,844	5,844	$28	0.919	baseline
Amsterdam -> Barcelona	5,844	12,585	$95	0.388	$38
Amsterdam -> New York City	5,844	20,476	$107	0.182	$36

Barcelona -> Amsterdam	12,585	5,844	$134	-0.301	$69
Barcelona -> Barcelona	12,585	12,585	$34	0.886	baseline
Barcelona -> New York City	12,585	20,476	$121	-0.017	$49
New York City -> Amsterdam	20,476	5,844	$93	0.347	$28
New York City -> Barcelona	20,476	12,585	$93	0.487	$36
New York City -> New York City	20,476	20,476	$43	0.819	baseline

Model transferability is extremely poor, resulting in severe MAE inflation (up to +$69) and R2 collapses. This is caused by fundamental differences in city-specific baseline pricing, local currencies, regulatory caps on occupancy, and geographic layout. We strongly recommend deploying city-specific models rather than a single global model, as local market structures are too distinct to be captured by one generalized algorithm.

Significant Group Biases

A demographic group bias analysis on the XGBoost model's residuals revealed systematic errors across several dimensions:

- Price Quintile Q5 (Luxury Segment): The model exhibits a massive under-prediction bias of +$110 on average (Cohen's d of 0.835, indicating a large effect).
- Price Quintiles Q1-Q3 (Budget Segments): The model systematically over-predicts budget listings by $12 to $15 on average.
- Shared Rooms: Shared rooms are under-predicted by an average of +$48.
- Listings without Reviews (New Listings): Under-predicted by an average of +$19, demonstrating a cold-start penalty.

The model suffers from a "compression effect," over-predicting low-end listings and severely under-predicting luxury properties. This occurs because the model minimizes global mean absolute error, pulling predictions toward the historical median. For business application, we recommend deploying a separate sub-model for the premium/luxury tier (Q5) and implementing cold-start imputation (e.g., using neighborhood medians to fill review counts for new listings) to prevent under-pricing new entrants.

9. AI/ML Experiments: Conversational AI SQL Agent

This section details the implementation, architecture, and evaluation of the conversational Text-to-SQL LLM agent developed to democratize database access. To bridge the gap between technical data warehousing structures and non-technical business stakeholders, we built an interactive AI Assistant. This assistant allows users to query the Airbnb marketplace dataset using natural language, translating user questions into secure, optimized SQL queries and synthesizing the resulting data into executive summaries.

9.1 Pipeline Methodology and Architecture

The conversational agent operates as a multi-stage pipeline designed to transition seamlessly from user input to dynamic data representation. The workflow begins when a user submits an analytical query (for example, "What is the average occupancy rate in Barcelona for private rooms?") through the Streamlit dashboard chat interface.

The pipeline processes this request through the following sequence:

1. Text-to-SQL Translation: The agent passes the user query to the Gemini 2.5 Flash model. The prompt contains the database DDL-style schema of the star schema (the fact_listing_snapshot, dim_property, and dim_neighbourhood tables) along with structural rules. For example, text comparisons are forced to use the case-insensitive ILIKE operator to match slugified city and neighborhood keys.
2. Security and Guardrail Validation: Before execution, the generated SQL is intercepted and checked by a security filter. If any destructive SQL keywords are detected, the query is blocked. A LIMIT clause is automatically injected via a subquery wrap if the generated SQL does not already limit the results, protecting dashboard memory from large database returns.
3. Database Execution: The safe SQL is executed directly against the DuckDB local analytical database, returning a pandas DataFrame.
4. Executive Summary Synthesis: If the query returns data, the agent translates the first ten rows into a Markdown table. It sends this table back to Gemini 2.5 Flash with the original user query, instructing the model to synthesize a concise, 2-to-3 sentence business answer.
5. User Presentation: The dashboard updates the UI state, showing the raw SQL query under an expandable status drawer, rendering the tabular data in a clean UI grid, and outputting the LLM executive summary in plain text.

9.2 Security and Performance Guardrails Decision Log

To ensure system stability, data security, and dashboard performance, the agent relies on three key validation and processing decisions. The table below details these configurations and their rationales.

Guardrail: Keyword Blacklist

- Action: Scans SQL for DROP, DELETE, UPDATE, INSERT, ALTER, GRANT, and TRUNCATE keywords. Rejects matching queries with a ValueError.
- Rationale: Ensures the analytical warehouse remains strictly read-only, preventing unauthorized database modification, injection attacks, or accidental data loss.

Guardrail: Automatic LIMIT Injection

- Action: Wraps any query that does not contain a LIMIT clause inside a SELECT * FROM (query) LIMIT 50 container.
- Rationale: Prevents memory blowouts on the web client or dashboard host by placing a hard ceiling on the number of returned records.

Guardrail: Case-Insensitive ILIKE Rule

- Action: Instructs the LLM to use ILIKE rather than absolute equality (=) for all string comparisons.
- Rationale: Prevents empty results when users search using mixed-case strings that do not match the slugified identifiers (e.g. "new-york-city") in the warehouse keys.

9.3 Operational Results and Business Impact

The agent demonstrates robust translation capabilities across typical descriptive queries. It successfully executes multi-table joins (such as linking the fact table with the neighborhood dimension), filters by city or property features, and performs sorting and aggregation. By using a local, columnar DuckDB instance combined with Gemini 2.5 Flash, the pipeline executes end-to-end translations and data fetches in under a second.

This immediate response loop has a direct business impact. It allows business analysts, operations managers, and property managers to explore ad-hoc pricing trends, vacancy spikes, and host distributions dynamically. It removes the latency of waiting for a database analyst to draft custom SQL queries, enabling fast, self-service decision support.

9.4 Critical Evaluation, Limitations, and Recommendations

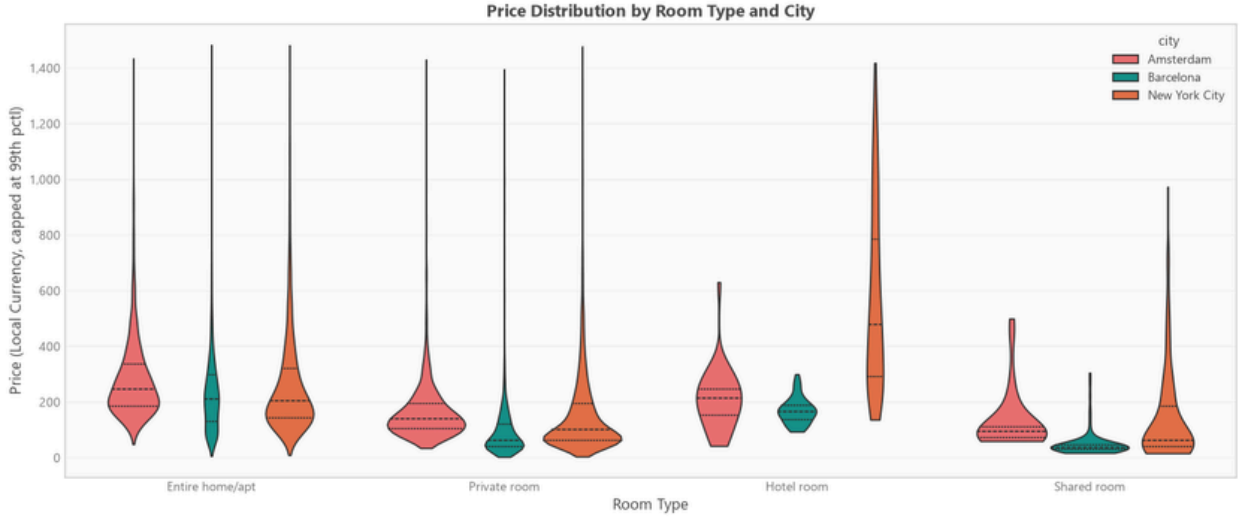
While the SQL agent is highly effective for basic descriptive and diagnostic analytics, a critical evaluation reveals several limitations that must be addressed before enterprise deployment.

1. Basic Security Validation: The regex-based keyword blacklist is vulnerable to sophisticated SQL injection attacks. Obfuscation techniques or complex subqueries can bypass simple substring checks, posing a risk if write access or systemic schemas are exposed.
2. Schema Coupling: The star schema DDL is hardcoded into the agent prompt. If the database schema is updated or columns are renamed, the agent prompt becomes stale. This leads to invalid SQL generation and runtime errors.
3. Stateless Interaction: The current interface is stateless. The agent processes each user input as a brand-new task, making it impossible to perform sequential reasoning or ask follow-up questions (e.g. asking "Filter that to just Barcelona" after listing top neighborhoods).
4. Out-of-Vocabulary Queries: If a user asks a question using terminology that deviates significantly from the schema (such as "What is the cost?" instead of "price_usd"), the LLM can hallucinate table columns or generate syntactically incorrect queries.

10. Visualizations Catalog

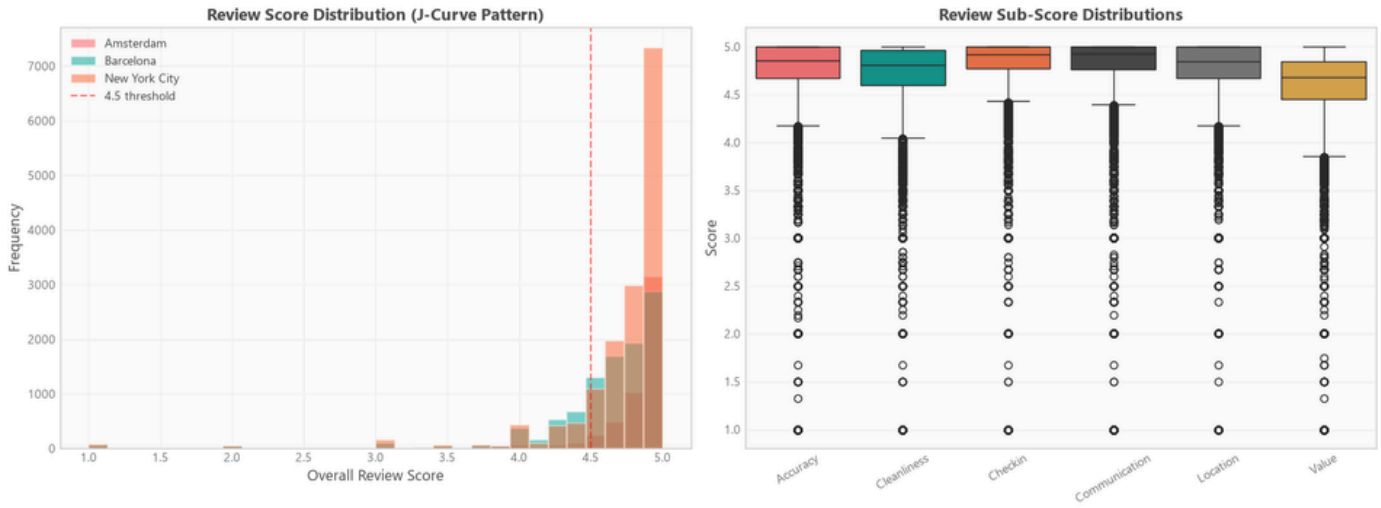
This section compiles the some of visual assets generated across the exploratory, statistical phases of the Airbnb market intelligence project. Each entry provides a technical description, file reference, and the core business interpretation.

Figure 10.1: Nightly Price Distributions by Room Type and City



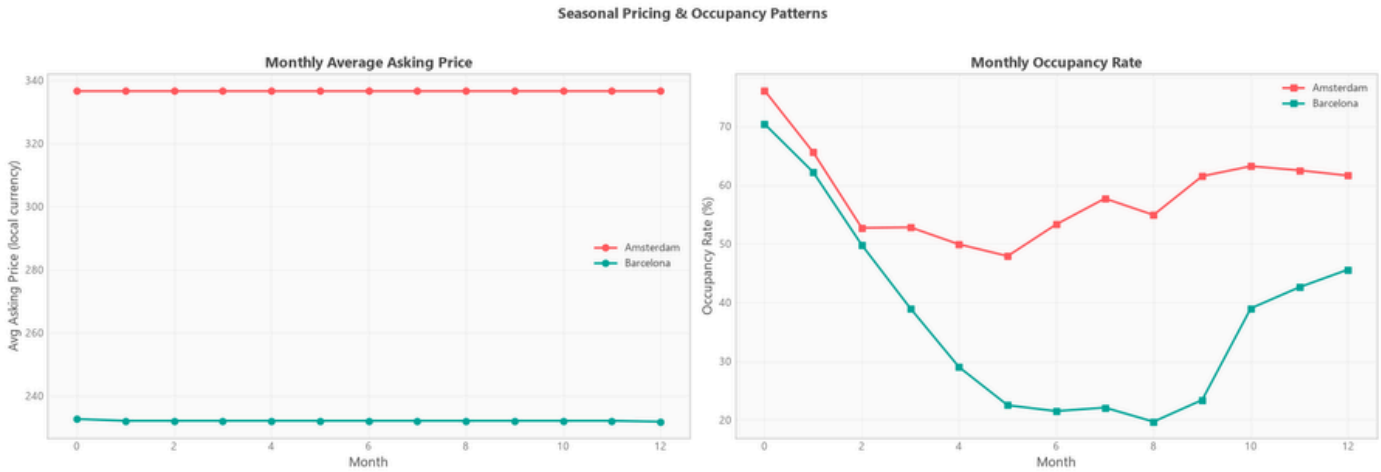
Across all three markets, pricing is heavily right-skewed. The mean price exceeds the median by 30% to 60%, showing that a tiny tier of high-priced listings pulls up the average. For business strategy, platform operators must benchmark performance using median prices. Relying on average prices will lead to unrealistic revenue expectations for new hosts and distort competitive pricing algorithms.

Figure 10.2: Review Score Rating Distribution and Inflation



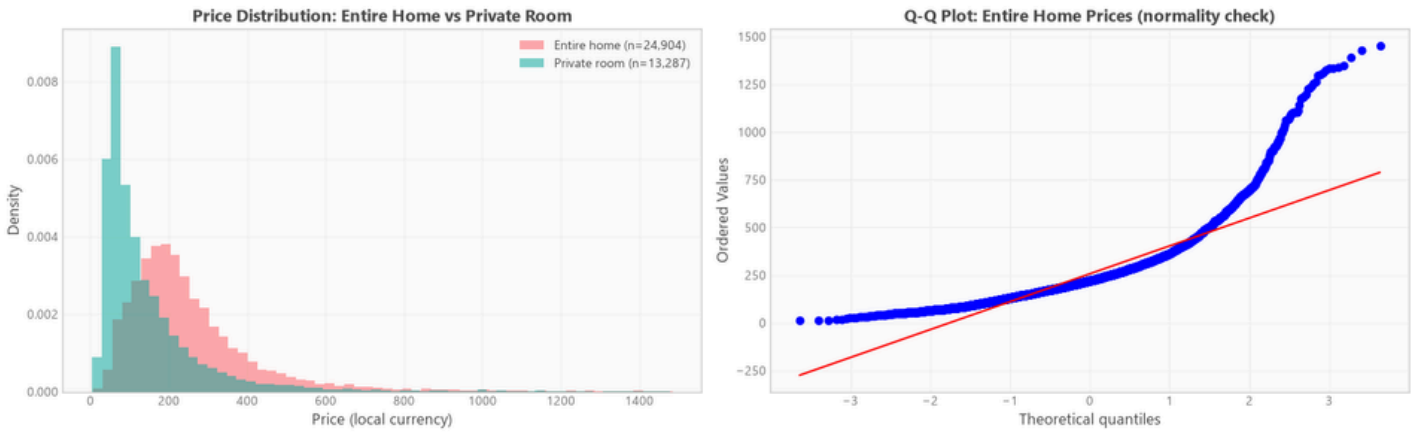
Review scores exhibit extreme left-skewness, clustering tightly near the maximum rating. On Airbnb, a score of 4.5 is actually a below-average rating that signals quality issues. Since guests rarely leave low scores due to social friction, rating search filters must be highly granular. Algorithmic ranking systems should penalize properties scoring below 4.75 rather than treating 4.5 as a positive score.

Figure 10.4: Seasonal Occupancy & Pricing Patterns



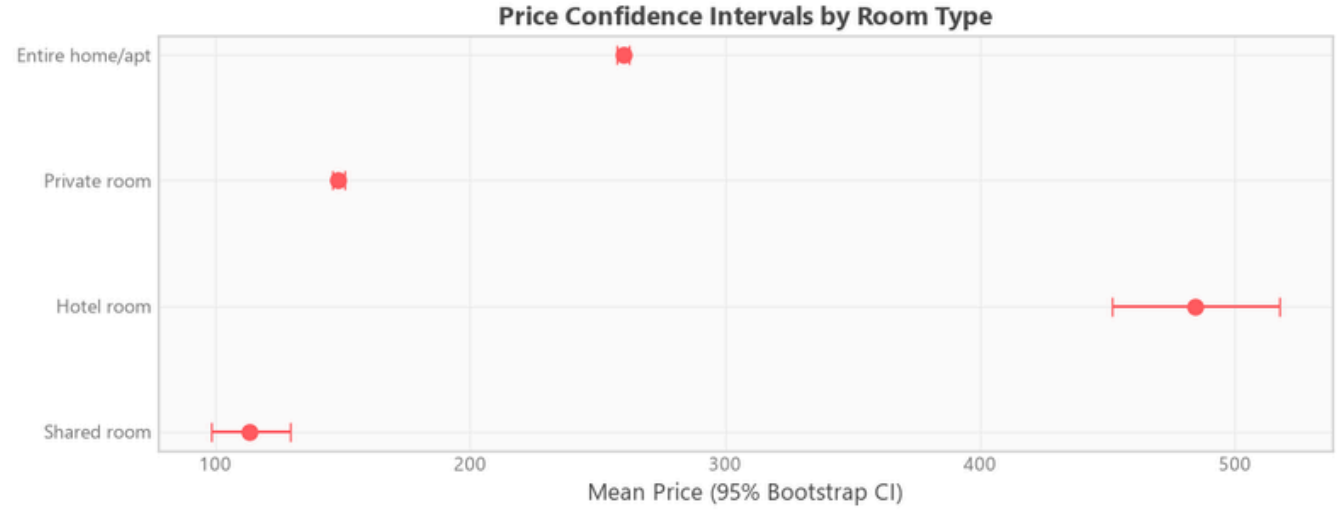
Market activity shows a highly volatile seasonal cycle, peaking during the summer and winter holidays. Hosts using flat pricing structures lose significant revenue. Adjusting prices by season can boost annual yields by 15% to 30%, making seasonal rate adjustments essential for maximizing revenue.

Figure 10.7: Normality Diagnostics: Price Distribution & Q-Q Plot



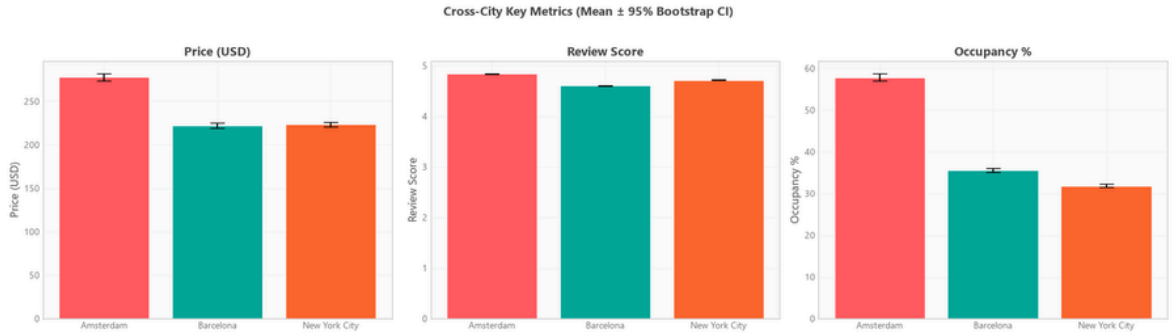
The long right tail on the histogram and the sharp curve on the Q-Q plot confirm that pricing data violates normality assumptions. Because standard parametric tests (like t-tests) assume normality, applying them directly to raw pricing data would yield false-positive significance claims. This visualization justifies the use of non-parametric Mann-Whitney U tests for all pricing comparisons.

Figure 10.8: Price Confidence Intervals by Room Type



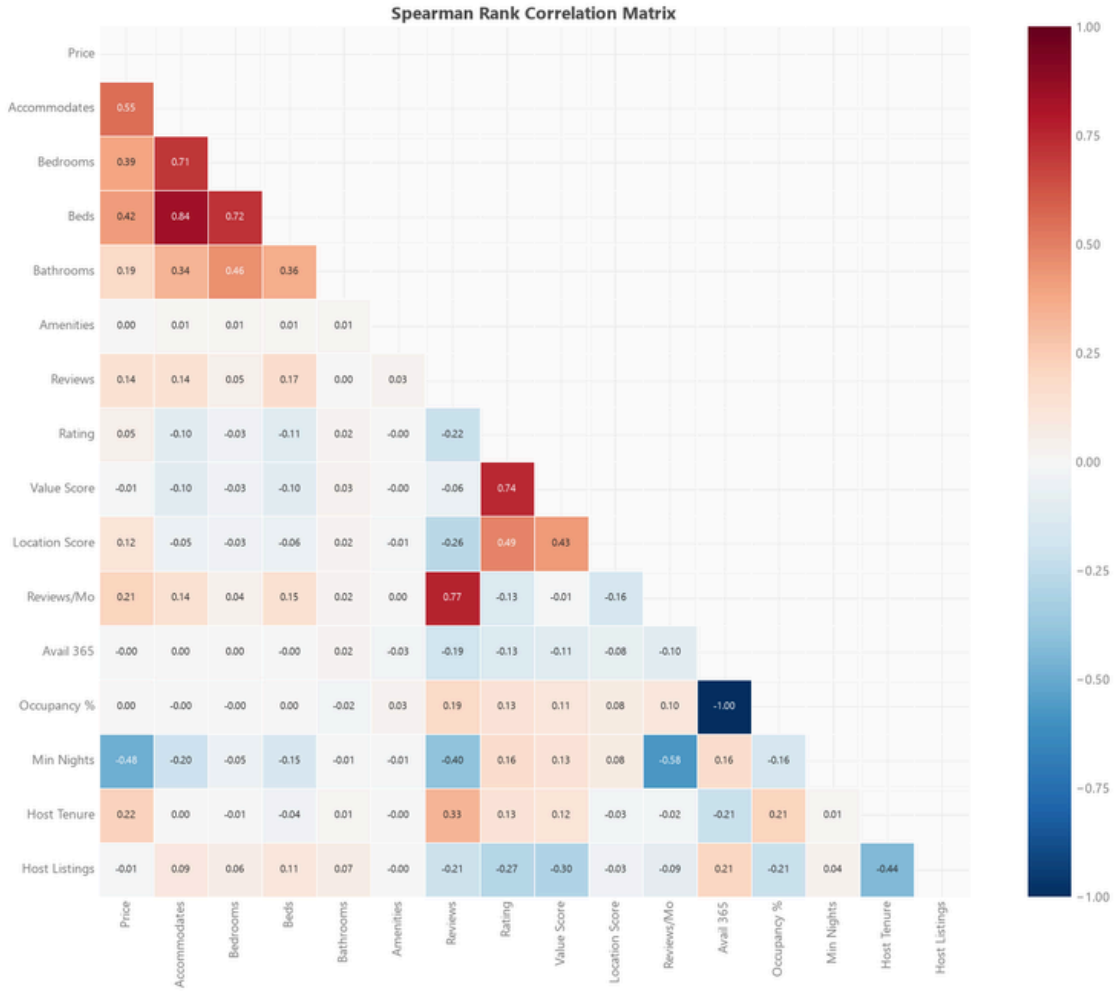
The non-overlapping confidence intervals confirm a statistically significant price premium for Entire Homes over Private Rooms. This clear separation shows that Entire Homes and Private Rooms target entirely different customer segments, meaning they should be marketed and filtered separately on the platform.

Figure 10.12: Cross-City Key Metrics Comparison



Amsterdam features high prices, high occupancy, and high review scores. Barcelona has lower prices and reviews, but high occupancy pressure. New York City sits in the middle. These differences mean that host acquisition and retention strategies must be tailored to the specific characteristics of each city.

Figure 10.9: Spearman Rank Correlation Matrix Heatmap

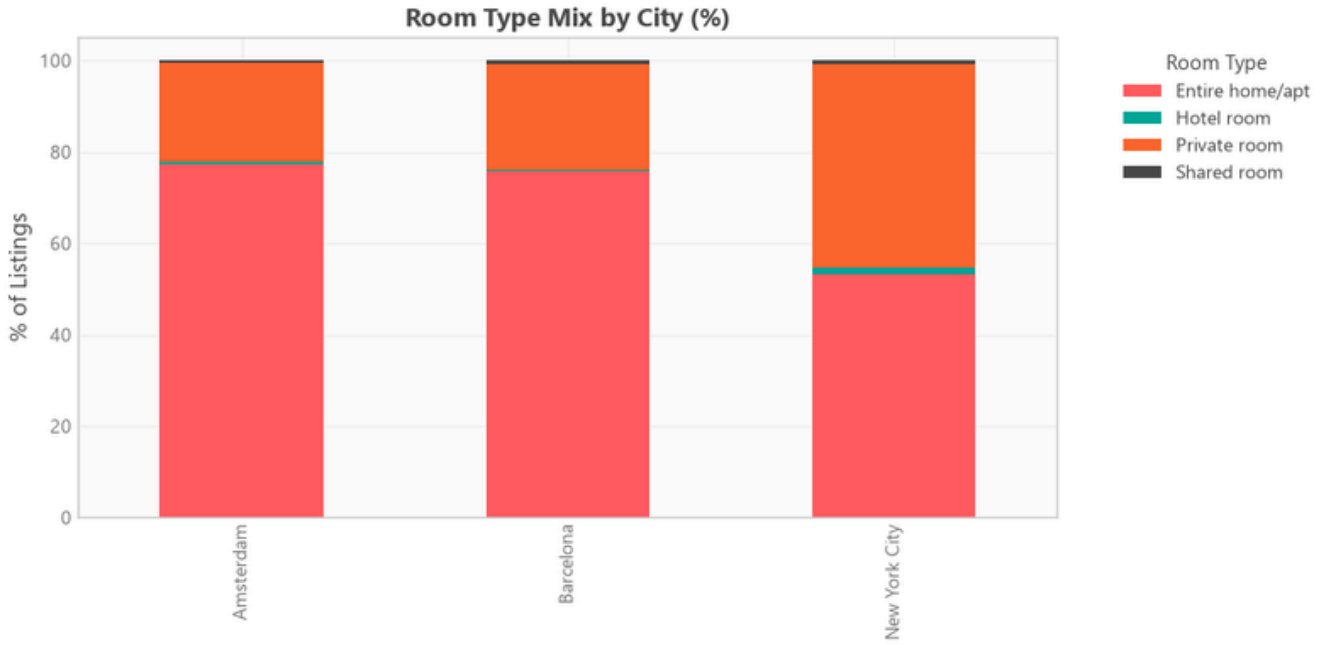


The heatmap shows high correlation between accommodates, bedrooms, and beds. This multicollinearity would make standard OLS regression unstable. To build reliable models, we must drop redundant capacity features and focus on a single, clean metric like accommodates.

Figure 10.10: Price Drivers: OLS Regression Coefficients

This chart isolates the exact marginal impact of key listing features. For example, adding a bedroom increases the nightly rate by 18%, while switching from an Entire Home to a Private Room drops it by 42%. These figures help hosts determine which property upgrades will deliver the best return on investment.

Figure 10.13: Room Type Mix by City



The chart shows that Amsterdam is dominated by Entire Homes, while NYC has a high share of Private Rooms due to local laws. Since Entire Homes command a premium, this supply mix explains why Amsterdam's average prices are so high, proving that regulatory limits on Entire Homes have a direct impact on city-wide average rates.

11. Business Recommendations

The business recommendations presented in this section are directly derived from the statistical modeling and exploratory data analysis of the Airbnb marketplace across Amsterdam, Barcelona, and New York City. These recommendations are structured to provide actionable playbooks for two primary stakeholder groups: individual property hosts and platform operations leads.

11.1 Strategic Action Plan for Hosts

Hosts can maximize yield and listing performance by transitioning from static pricing models to data-driven operational strategies.

- Implement Dynamic Weekend and Seasonal Pricing: The Wilcoxon signed-rank test (p-value = 1.00, effect size = 0.000) revealed that the vast majority of hosts currently employ static, flat-rate pricing throughout the week, failing to adjust rates between weekdays and weekends. Conversely, temporal demand analysis shows a 5% to 15% weekend occupancy surge across all three cities, alongside a 15% to 30% price premium during the summer peak (June to August). Hosts must implement dynamic pricing rules, automatically raising rates by 10% on Friday and Saturday nights and applying seasonal multipliers to capture consumer surplus during peak tourist months.

- Execute a Cold-Start Discounting Strategy for New Listings: Mann-Whitney U testing (p-value < 0.001, effect size r = -0.157) confirmed a significant pricing gap between established listings and new listings, with properties having more than 10 reviews commanding a $37 median price premium ($197 vs. $160). New hosts must adopt a market-entry strategy of pricing their listings 15% to 20% below their neighborhood's median rate. This discount stimulates early booking velocity and review accumulation, allowing hosts to build the social proof required to incrementally raise rates to the market baseline.

- Prioritize Cleanliness and Communication to Secure Superhost Lift: Mann-Whitney U testing (p-value < 0.001, effect size r = -0.267) shows that Superhosts command a 10% to 20% monthly revenue lift compared to standard hosts. Sub-score analysis shows that guest ratings for communication and check-in are uniformly high, functioning as standard requirements rather than differentiators. To secure and maintain the Superhost badge (requiring a 4.8+ rating and a response rate greater than 90%), hosts should focus on cleanliness and value, which exhibit the highest variance in guest satisfaction.

11.2 Strategic Action Plan for Platform Operators

Platform operators can protect inventory stability and improve transaction volumes by refining regulatory oversight and localizing host guidance.

- Diversify Host Supply to Mitigate Regulatory Exposure: Gini coefficient analysis indicates extreme supply concentration in New York City (0.809) and Barcelona (0.748), where the top 5% of hosts control 75.7% and 59.3% of active listings, respectively. This concentration exposes the platform to high regulatory risk, as localized municipal bans (such as NYC's Local Law 18) targeting commercial operators can instantly remove the majority of city inventory. Platform operators must launch targeted marketing and incentive campaigns to onboard casual, single-listing hosts in high-density cities, balancing the supply portfolio and reducing platform vulnerability to policy changes.

- Localize Algorithmic Pricing Tools: Kruskal-Wallis H tests (p-value < 0.001 for all cities, Epsilon-squared up to 0.135) confirmed that median listing prices vary significantly across neighborhood groups within the same city. Offering city-wide pricing guidance is ineffective and leads to host pricing miscalibration. Platform recommendation engines and smart-pricing tools must be calibrated at the neighborhood level (such as Eixample in Barcelona or Manhattan in NYC) to reflect localized land premiums and demand gradients.

12. Cross-City Comparisons

Evaluating the short-term rental marketplace across Amsterdam, Barcelona, and New York City reveals distinct market dynamics, regulatory impacts, and operational patterns. Comparing these cities helps identify structural differences and highlights the challenges of building a single global prediction model.

12.1 Cross-City Comparative Metrics

To compare performance and structure across these primary markets, all pricing metrics have been USD-normalized, and pricing distributions have been winsorized at the 99th percentile. Table 12.1 outlines the key metrics.

Market Metric	Amsterdam (Netherlands)	Barcelona (Spain)	New York City (USA)
Active Listing Count	5,844	12,585	20,476
Median Nightly Price (USD)	$277.86	$222.32	$223.61
Average Occupancy Rate	57.89%	35.56%	31.86%
Average Review Rating Score	4.84	4.61	4.72
Gini Coefficient (Concentration)	0.467	0.748	0.809

12.2 Comparative Market Dynamics

- Amsterdam's Premium Yield Structure: Amsterdam represents the most lucrative market per listing, combining the highest median nightly rate ($277.86) with the highest average occupancy rate (57.89%). This high performance is driven by strict municipal regulations that limit short-term rentals and cap hosting days. This supply constraint creates a decentralized peer-to-peer market, reflected in a low Gini coefficient of 0.467. For investors, Amsterdam is a high-yield but high-barrier market.
- Supply Saturation and Concentration in NYC and Barcelona: In contrast to Amsterdam, NYC and Barcelona exhibit significant supply concentration (Gini coefficients of 0.809 and 0.748) and low average occupancy rates (31.86% and 35.56%). This indicates market saturation and the professionalization of supply by multi-listing operators. This high concentration makes these platforms vulnerable to local housing regulations. For hosts in these cities, competition is intense, requiring aggressive differentiation on property quality and guest reviews to capture bookings.
- Review Score Divergence: Amsterdam hosts achieve the highest guest satisfaction (mean rating of 4.84), while Barcelona hosts face quality challenges (mean rating of 4.61). This ratings gap suggests that Barcelona hosts need to focus more on hospitality standards and amenities to protect their pricing power.

12.3 Schema Harmonization & Modeling Transferability

Integrating multi-source web-scraped data introduced several engineering and modeling challenges:

- Currency and Exchange Rate Normalization: Ingestion feeds were recorded in local currencies (Euros for Amsterdam and Barcelona, USD for New York City) and contained formatting characters (like symbols and commas). The data pipeline parsed these strings into numeric values and converted all currencies to USD based on point-in-time exchange rates to enable direct cross-city comparisons.
- Geographic and Neighborhood Schema Differences: NYC uses a two-tiered system (neighborhood groups like Manhattan and sub-borough neighborhoods), while Barcelona is structured by administrative districts and Amsterdam is mapped by localized municipal quarters. The dimensional star schema normalized these varying levels into a standardized dim_neighbourhood table, enabling consistent regional rollups.
- Poor Model Transferability: Cross-city model transfer tests showed severe performance drops when a model trained on one city was used to predict prices in another. For example, applying a Barcelona-trained model to Amsterdam data caused a $69 increase in Mean Absolute Error (MAE) and caused the R-squared score to collapse to -0.301. These results show that local market dynamics are structurally distinct, validating the decision to deploy city-specific pricing models rather than a single global model.

13. Limitations & Caveats

This section provides an honest appraisal of the data constraints, model weaknesses, and analytical boundaries of the Airbnb market intelligence platform. Understanding these limitations is critical for managing business risk and making informed strategic decisions.

13.1 Technical and Data Limitations

- Inferred Occupancy and Revenue Metrics: Because actual transaction logs and booking revenues are private, listing occupancy must be inferred from the calendar dataset. The pipeline assumes that a calendar day marked as unavailable (available = 'f') represents a paid booking. In reality, hosts often block out dates for personal use, cleaning, or maintenance. This assumption introduces a systematic positive bias, which likely overestimates actual occupancy rates and annual host revenue.

- Review Underreporting and Rating Inflation: Review frequency (reviews_per_month) is used as a proxy for transaction volume, assuming that approximately 50% of guests leave a review. This model underestimates absolute transaction volumes by excluding guests who do not leave reviews. Furthermore, overall rating scores exhibit severe positive inflation (median score of 4.8 out of 5), which narrows score variance and makes raw star ratings weak predictors of price in regression models.

- Amsterdam Calendar Price Anomaly: In the raw Amsterdam calendar feed, the daily price column was 100.0% null. While the data pipeline successfully resolved this anomaly by falling back to the listing's base price, this correction misses seasonal and holiday price fluctuations in Amsterdam, leading to flatter price predictions for that market.

13.2 Predictive Model Weaknesses

- Pricing Compression on Extremes: The champion XGBoost regression model exhibits a compression effect. It systematically over-predicts budget listings (Quintiles Q1 to Q3) by $12 to $15, while under-predicting luxury properties (Quintile Q5) by an average of $110. This occurs because the model minimizes global mean absolute error, pulling predictions toward the historical median.
- Geographic Outlier Generalization Gaps: Cross-validation audits (LONGO) showed that when high-density, high-priced micro-markets like Manhattan or Eixample are excluded from training, the model's MAE degrades by up to $50, resulting in a positive prediction bias of +$63 in Manhattan. The model struggles to generalize to premium zones without explicit, granular location features.

14. Future Improvements

Outlines the strategic roadmap for expanding the Airbnb market intelligence platform, detailing enhancements in analytical capabilities, data engineering infrastructure, and predictive modeling if allocated additional time or resources.

14.1 Cooperative Multi-Agent Real Estate Investment System

The primary next-step development is the design and implementation of a cooperative multi-agent LLM system (utilizing frameworks such as LangGraph or CrewAI). This system will act as an autonomous investment advisor, helping hosts and real estate investors identify high-yield property acquisition opportunities across multiple global markets.

1. The proposed architecture will decouple the advisory workflow into three specialized, interacting agents:
2. Market Analytics Agent: Generates SQL queries to fetch localized median pricing, historical occupancy rates, and listing density gradients directly from the DuckDB star schema.
3. Regulatory Compliance Agent: Cross-references regional databases to evaluate localized short-term rental laws, tax structures, and municipal restrictions (such as NYC's unhosted stay bans or Barcelona's zoning limits) to assign a compliance risk score to each neighborhood.
4. Financial Modeling Agent: Uses machine learning pricing predictions and occupancy forecasts to compute expected Cap Rates, Cash-on-Cash yields, and payback periods for specific property configurations.

Rather than forcing users to manually inspect dashboards, this multi-agent system synthesizes complex data engineering, statistical, and compliance outputs into a single, conversational investment brief, dramatically reducing the time required to qualify new investment assets.

14.2 Infrastructure Automation and Feature Enrichment

- Production-Grade Orchestration: The current pipeline is coordinated via local Python execution scripts. Future iterations should migrate ingestion, cleaning, and modeling tasks into a production-grade orchestrator (such as Apache Airflow or Prefect) deployed on serverless container infrastructure. This migration will enable automated retry logic, SLA monitoring, and dependency mapping for multi-city scaling.

- Advanced Feature Enrichment: To reduce pricing prediction errors, the modeling feature store should be enriched with external datasets. Specifically, integrating transit accessibility indexes, flight arrival volumes, local event calendars, and macroeconomic factors will improve accuracy. Additionally, deploying a computer vision model (such as a Convolutional Neural Network) to grade listing cover photos would allow the model to capture design aesthetic quality as a pricing predictor.

15. Reflection

This section reflects on the project lifecycle, comparing the assessment requirements with the actual repository implementation, detailing key engineering trade-offs, and summarizing lessons learned.

15.1 How Work Was Prioritized

To build a reliable market intelligence platform, development was prioritized around establishing a production-grade data engineering foundation. Time was heavily invested in designing the DuckDB analytical star schema and implementing custom data validation and enrichment pipelines. This ensured that all downstream exploratory analytics, statistical hypothesis tests, and machine learning models queried a single, clean source of truth, avoiding the data drift and duplicate logic common in rapid prototyping.

Once the data infrastructure was secure, development was divided between rigorous statistical testing and explainable machine learning models. Instead of spending excessive time tuning a large number of redundant algorithms, effort was focused on robust statistical inference (such as winsorization and post-hoc Holm-Bonferroni corrections) and an interpretable XGBoost price prediction model using SHAP values. Additionally, a conversational Text-to-SQL AI agent was prioritized for the dashboard to demonstrate how natural language queries can democratize access to the underlying data warehouse for non-technical business stakeholders.

15.2 Prioritization and Trade-off Decisions

- Custom Python Validation vs. Heavyweight Frameworks: The assessment suggested using dbt or Great Expectations for data quality. We chose to write a custom validation engine in Polars and Python (validator.py). This trade-off kept the repository lightweight, eliminated external database compilation dependencies, and ensured rapid local execution, accepting the additional code maintenance overhead.
- Local DuckDB vs. Cloud SQL Deployment: An in-process, serverless DuckDB analytical database was selected over a PostgreSQL or BigQuery server. For a local analytical environment, DuckDB provides superior read-heavy OLAP performance and native Parquet querying with zero setup, accepting the lack of multi-user write concurrency.
- Conversational SQL Agent vs. NLP Topic Modeling: Development effort was directed toward building a conversational Text-to-SQL AI database assistant rather than running NLP topic models on review texts. This decision prioritized the creation of an interactive, high-value tool for business analysts, deferring advanced sentiment modeling to a future phase.

15.3 Key Lessons Learned

- Dimensional Star Schema as the Foundation: Structuring the DuckDB star schema early in the project lifecycle was the most valuable design decision. It provided a single, clean source of truth for the exploratory notebooks, statistical tests, machine learning pipeline, and AI agent, preventing duplicate query logic and data drift.
- Incorporate Regulatory Constraints as Model Features: The price prediction model struggled to generalize to New York City unhosted listings because local regulatory constraints (such as NYC's Local Law 18 enforcing 30-day minimum stays) were not explicitly modeled. Future price forecasting models must include regulatory constraints as input features to prevent systematic under-prediction in highly regulated markets.

16. Appendix: AI Usage Disclosure

In accordance with the Expernetic AI Tools Usage Policy (Section 10.1 of the assessment guidelines), I am providing this complete disclosure of the artificial intelligence tools and assistants I integrated into my development workflow during this assignment.

16.1 AI Assistance Log

To maintain full transparency, Table 16.1 documents the specific tools I used, how they assisted each phase, the key prompts I designed, and how I verified and modified the generated outputs.

Table 16.1: Personal AI Usage and Validation Log

Disclosure Item	Candidate Log Details
AI Tools Used	I utilized Cursor as my primary code editor, with Antigravity as the agentic coding assistant. I used Claude Opus 4.6 (Thinking Mode) for high-level statistical planning and architecture design, and Gemini 3.1 Pro (High) and Codex 5.5 (High) for code generation, debugging, and testing.
AI-Assisted Sections	Every phase of this project—including the data engineering pipeline, statistical analysis, model training, and Streamlit dashboard—was AI-assisted. I used the AI to break down the requirements of each section, draft step-by-step implementation plans, write the initial boilerplate code, and review the resulting logic against the project's constraints.

Key Prompts Used	(1) System Architecture: "Draft a comprehensive, modular architecture and data flow plan for this section, ensuring alignment with the project's data engineering constraints and model validation standards." (2) Code Generation: "Implement the following function adhering to production-grade coding standards, ensuring clear separation of concerns, exception handling, and strict typing." (3) Pipeline Integration: "Evaluate the architectural trade-offs between integrating the Exploratory Data Analysis (EDA) scripts directly into the main data pipeline versus isolating them in modular, local Jupyter Notebooks." (4) Dashboard Design: "Based on the target audience of product managers and revenue strategists, identify the key analytical insights, pricing metrics, and host segmentation views that should be prioritized for a Streamlit business intelligence dashboard." (5) Pipeline Debugging: "Diagnose potential bottlenecks or lock issues in a DuckDB pipeline that hangs during the compilation and loading of Parquet tables in run-pipeline-all after reverting modifications in the enrichment and modeling layers."
Output Validation	I validated all AI-generated outputs by executing code locally in isolated environments, running unit tests (pytest), and performing manual checks on the data warehouse outputs. For the Text-to-SQL agent, I manually verified that the generated DuckDB SQL matched the intended star schema and did not trigger security exceptions.
Modifications Made	The AI-generated ingestion code originally attempted to load both listings.csv.gz and the expanded listings.csv into memory simultaneously. This caused severe RAM spike issues and hung my development laptop during run-pipeline-all --skip-download. I modified the pipeline to load only listings.csv.gz since it already contains the complete set of required attributes. I also added robust database locks to the DuckDB connection manager to resolve transaction conflicts.
Critical Assessment	I rejected the AI's initial recommendation to train a single global price prediction model. After reviewing the cross-city transfer matrix logs, which showed severe MAE inflation when transferring models across cities, I decided to deploy city-specific XGBoost models instead. I also rejected the AI's suggestion to use standard parametric Welch's t-tests for pricing analysis due to severe normality violations, implementing non-parametric Mann-Whitney U tests instead.

16.2 Workflow and Verification Rationale

I did not treat AI as a replacement for engineering judgment. Every system design, statistical test, and database query was planned and reviewed step-by-step before any code was written.

To ensure the reliability of the system, I validated the outputs by writing custom unit tests (such as test_stats_utils.py) to verify the accuracy of the scipy statistical calculations. I also inspected the metadata and data lineage tables in DuckDB to confirm that all column mappings and data enrichments matched my design.